

#HW 7

● Graded

Student

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Total Points

76 / 100 pts

Question 1

(no title)

9 / 10 pts

– 0 pts Correct

– 2 pts ΔT is same in both rankine and fahrenheit scales.

– 2 pts 6 inches is 0.5 ft

– 5 pts $Q = K A \Delta T / L$.

– 3 pts check the calculation properly.

✓ – 1 pt total heat loss in 10 hours not calculated

– 10 pts not answered

Question 2

(no title)

13 / 15 pts

– 0 pts Correct

✓ – 2 pts $Q_{\text{total}} = Q_{\text{convection}} + Q_{\text{radiation}}$; step missed

– 2 pts check the calculation properly

– 15 pts question hasn't been answered

Question 3

(no title)

7 / 15 pts

– 0 pts Correct

✓ – 3 pts $W = m (g/gc)$

✓ – 3 pts $m = PV/RT$

✓ – 2 pts check the units/ calculation properly

– 10 pts didnt answer the question properly

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts calculation error

- 5 pts $Q = m C_p \Delta T$

- 5 pts incomplete calculation

Question 5

(no title)

12 / 15 pts

- 0 pts Correct ; 634 K or 361 C

✓ - 3 pts calculation error

Question 6

(no title)

12 / 15 pts

- 0 pts Correct

- 5 pts incomplete solution

✓ - 3 pts Calculation mistake/ error.

Question 7

(no title)

8 / 15 pts

- 0 pts Correct

✓ - 3 pts calculation error in work done

✓ - 2 pts calculation error in Q

✓ - 2 pts part b calculation error/ not completed.

$$1) \dot{Q} = -KA \frac{dT}{dx} \quad A = (20\text{ ft})(10\text{ ft}) = 200\text{ ft}^2 \quad Q = -(10\text{ hr})(0.04 \frac{\text{BTU}}{\text{hr ft}^2 \text{ } ^\circ\text{R}})(200\text{ ft}^2)(-28) / (0.5)$$

$$K = 0.04 \frac{\text{BTU}}{\text{hr ft}^2 \text{ } ^\circ\text{R}} \quad t = 10\text{ hr} \quad \boxed{Q = 448 \text{ BTU}}$$

$$T_1 = 68^\circ\text{F} = 527.67^\circ\text{R} \quad dx = 6\text{ in} = 0.5\text{ ft}$$

$$T_2 = 40^\circ\text{F} = 499.67^\circ\text{R} \quad \Delta T = 499.67 - 527.67 = -28^\circ\text{R}$$

$$2) T_b = 47^\circ\text{C} = 320\text{ K} \quad \text{Convection:} \quad \dot{Q} = hA(T_b - T_f)$$

$$T_f = 27^\circ\text{C} = 300\text{ K} \quad \dot{Q} = h(T_b - T_f) \quad \text{radiation:} \quad \dot{Q} = \epsilon \sigma A(T_b^4 - T_f^4)$$

$$h = 10 \frac{\text{W}}{\text{m}^2 \text{ K}} \quad \dot{Q} = (10 \frac{\text{W}}{\text{m}^2 \text{ K}})(320 - 300\text{ K}) (\frac{1\text{ K}}{1000\text{ W}}) \quad \frac{\dot{Q}}{A} = (0.93)(5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{ K}^4})(320^4 - 300^4)(\frac{1\text{ K}}{1000\text{ W}})$$

$$\epsilon = 0.93 \quad = 0.2 \text{ kW/m}^2 \quad = 0.126 \frac{\text{W}}{\text{m}^2}$$

$$G = 5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{ K}^4}$$

$$3) p = 10\text{ atm} = 14.7\text{ psi} \quad PV = nRT$$

$$V = (110\text{ ft})(60\text{ ft})(25\text{ ft}) = 165000\text{ ft}^3 \quad m = \frac{PV}{RT}$$

$$T = 70^\circ\text{F} = 529.67^\circ\text{R} \quad m = (14.7\text{ psi})(165000\text{ ft}^3) / ((529.67^\circ\text{R})(10.732 \frac{\text{ft}^3\text{ psi}}{\text{lb}^\circ\text{R}}))$$

$$m = 4914 \times 10^4 \text{ lb} \quad (32\text{ ft/s}^2) = 1.573 \times 10^6 \text{ lb}$$

$$4) Q = mc\Delta T$$

$$m = 2\text{ kg} \quad Q = (2\text{ kg})(1.002 \frac{\text{kJ}}{\text{kg K}})(769.79 - 384.89)$$

$$c = \frac{K}{K-1} = \frac{1.35(8.314 \frac{\text{J}}{\text{kmol K}})(\frac{1\text{ kmol}}{32\text{ kg}})}{1.35 - 1} = 1.002 \frac{\text{kJ}}{\text{kg K}} \quad \boxed{Q = 771.33 \text{ kJ}}$$

$$PV = nRT$$

$$T_1 = \frac{PV}{nR} = (1\text{ bar})(10^2 \frac{\text{kg}}{\text{m}^3})(2\text{ m}^3) / (2\text{ kg})(8.314 \frac{\text{J}}{\text{kmol K}})(\frac{1\text{ kmol}}{32\text{ kg}}) = 384.89\text{ K}$$

$$T_2 = (1\text{ bar})(10^2 \frac{\text{kg}}{\text{m}^3})(4\text{ m}^3) / (2\text{ kg})(8.314 \frac{\text{J}}{\text{kmol K}})(\frac{1\text{ kmol}}{32\text{ kg}}) = 769.79\text{ K}$$

$$5) \dot{Q} = 425\text{ W}$$

$$Q = (425\text{ W})(10\text{ min})(\frac{60\text{ s}}{1\text{ min}}) = 255\text{ kJ} \quad Q = mc(T_f - T_o)$$

$$m = 5\text{ kg} \quad T_f = \frac{Q}{mc} + T_o$$

$$c = \frac{K}{K-1} = \frac{1.35(8.314 \frac{\text{J}}{\text{kmol K}})(\frac{1\text{ kmol}}{32\text{ kg}})}{1.35 - 1} = 1.002 \frac{\text{kJ}}{\text{kg K}} \quad T_f = \frac{255\text{ kJ}}{5\text{ kg}(1.002 \frac{\text{kJ}}{\text{kg K}})} + 703\text{ K}$$

$$T_1 = 430^\circ\text{C} = 703\text{ K} \quad \boxed{T_f = 753.90\text{ K}}$$

$$6) Q = 1.7\text{ BTU} \quad PV = nRT$$

$$T_1 = 560^\circ\text{R} \quad m = \frac{PV}{RT} = (18\text{ psi})(0.29\text{ ft}^3) / ((560^\circ\text{R})(10.732 \frac{\text{ft}^3\text{ psi}}{\text{lb}^\circ\text{R}}))$$

$$V_1 = 0.29\text{ ft}^3 \quad m = 19.60\text{ lb}$$

$$p_1 = 18\text{ psi} \quad Q = mc\Delta T$$

$$c = 0.171 \frac{\text{BTU}}{\text{lb } ^\circ\text{R}} \quad T_f = \frac{Q}{mc} + T_o$$

$$T_f = \frac{1.7\text{ BTU}}{(19.6\text{ lb})(0.171)} + 560 = 561^\circ\text{R}$$

$$7) \quad p_1 = 1.7\text{ bar} \quad PV^{1.25} = \text{constant} \quad W = \int_{V_1}^{V_2} \frac{1.25}{V^{1.25}} dV$$

$$T_1 = 280\text{ K} \quad p_1 V_1^{1.25} = p_2 V_2^{1.25} \quad W = - (1.31)(V^{0.25}) / 0.25 \big|_{0.262}^{0.29}$$

$$p_2 = 11\text{ bar} \quad V_2 = (p_1 V_1^{1.25} / p_2)^{1/1.25} \quad = -9.65 \times 10^{-3} (10^2 \frac{\text{kg}}{\text{m}^3})$$

$$V_1 = 0.262\text{ m}^3 \quad V_2 = 0.029\text{ m}^3 \quad W = 0.965\text{ kJ}$$

$$\text{constant} = (1.7)(0.262)^{1.25} = 0.13 \quad PV = nRT$$

$$m = \frac{PV}{RT} = (1.7\text{ bar})(0.262\text{ m}^3) / ((280\text{ K})(8.314 \frac{\text{J}}{\text{kmol K}})(\frac{1\text{ kmol}}{44.01\text{ kg}}))$$

$$m = 0.35\text{ kg} \quad \frac{.7}{280} = \frac{11}{T_2} \quad T_2 = 4400\text{ K}$$

$$Q = mc\Delta T$$

$$Q = (0.35\text{ kg})(4400 - 280)(.287)$$

$$\boxed{Q = 413\text{ kJ}}$$